A Hybrid CNN–BiLSTM–Attention Model for Noise-Resilient Dog Emotion Recognition Using Augmented Mel-Spectrogram Data

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Abstract—**The paper introduces a hybrid Deep Learning-based classifier designed to efficiently identify and categorize different emotional states in dogs. The hybrid classifier is built by sequentially combining three well-known Deep Learning architectures: CNN, Bi-LSTM, and Self-Attention modules. In order to extract the most relevant features from the acoustic signals of dog emotions, two signal analysis techniques: Spectrogram and Log-Mel-Spectrogram are utilized. Furthermore, the raw dog-emotion signals are mixed with real-time noises, including air conditioning, drilling, jackhammer, and engine idling, in order to measure the robustness of the classifier under noisy conditions. The real-time noise samples are drawn randomly from the UrbanSound8K open-source dataset. Optionally, a denoising scheme based on Wiener filtering in the frequency domain is employed on the noisy dog-emotion data. The resulting raw data, noisy data, and denoised data are applied to the hybrid classifier to determine the classifier's effectiveness. The types of dog voice emotion classes considered for classification include bark, growl, grunt, and whining. The proposed hybrid model applied to the dog voice emotion dataset with log-mel spectrogram features of voice samples produced a validation accuracy of 92%, 99%, and 97% on the clean, noisy, and denoised dog-emotion samples**

Index Terms—**Dog-voice Emotion, Convolutional Neural Network, CNN-BiLSTM, CNN-BiLSTM-Attention**

I. Introduction

HUMANS share a special bond with dog. Historians documented there bonding evolved approximately 15000 years ago. Certain benefits offered by the adoption of domestic dog alongside humans includes increased exercise, stress reduction, healthy cardiovascular activities, better immune support etc. Boris M Levinson, popularly known as the father of animal assisted therapy, made a significant influence in the use of dogs in clinical settings. He discovered that an introvert or mentally troubled child would interact kindly with an unfamiliar dog, and their communication continued for a while. In addition, dogs offer meaningful therapeutic benefits to older adults facing social isolation or physical challenges. Interaction with dogs has been shown to improve verbal expression, support the ability to articulate thoughts, and help maintain cognitive sharpness over time. Communicating with a dog often reduces the depressive symptoms observed in older adults [1].

Young single adults facing the growing problem of loneliness and the uncertainties of early career life are increasingly turning to pets for emotional support. Research published in [2] highlights that, for individuals aged 18–26, pets function as "social catalysts," offering a safe and non-judgmental environment to develop emotional regulation skills. For single individuals, pets often become meaningful companions, helping to ease social isolation while encouraging a sense of connection with the wider community.

A notable proportion of university students-approximately 10% exhibit symptoms associated with borderline personality disorder (BPD), a condition characterized by disturbances in self-identity, low self-esteem, and difficulties in emotional regulation. Caring for a pet can foster a sense of purpose and enhance personal value, which may be particularly beneficial for individuals experiencing BPD-related symptoms [3]. In this context, dog owners have been found to report lower levels of anxiety, depression, and somatization, along with higher self-esteem compared to students without pets

The work in [4] reviews emotion recognition in companion animals from an engineering viewpoint. It models emotional states as pattern recognition problems using vocal, facial, and behavioral cues. The study emphasizes signal processing–based feature extraction and classical machine learning methods, with discussion on emerging deep learning approaches. Challenges such as inter-animal variability, noise, and limited labeled data are highlighted. This work motivates robust and data-driven emotion recognition frameworks.

In most cases, pet owners maintain a caring and compassionate relationship with their dogs, characterized by kindness and positive interaction. However, on rare occasions, miscommunication between humans and dogs may occur, which can lead to frustration and, in some instances, inappropriate or abusive behaviour. Recognizing and addressing these situations is important to ensure animal welfare and to promote healthier human–animal relationships. In this context, automatic emotion recognition from dog vocalizations emerges as both a challenging and compelling area of study. Accurate interpretation of dog vocal signals can help reduce human–dog miscommunication and support more empathetic interactions. Accordingly, this paper introduces a novel hybrid deep learning–based classifier designed to automatically recognize and classify different dog vocalization samples under clean, noisy, and denoised conditions. Some of the key objectives of the proposed system includes.

- To design an efficient voice recognition model to classify dog-vocal emotions
- To compare the effectiveness of classifier using different feature extraction techniques
- To compare state of art Deep learning classifiers with the proposed hybrid classifier on dog emotion dataset

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- To add real time noise to the dog-voice samples and measure the effectiveness of the proposed classifier on the noisy voice samples.

II. Literature Review

Yang et al. [5] employed Synchrosqueezing STFT to extract the time frequency features of dog barks. Recognition model was based on MAMBA for global modelling of long length sequence features to effectively process time-frequency features and accurate identification of their variations. Three different datasets namely, self-constructed dog emotion sound dataset, UrbanSound8K, IEMOCAP were considered. The systems recognition accuracies on the above dataset are 91.97%, 95.36% and 67.25% respectively.

Karaslan et al. [6] aimed at comparing various CNN architectures namely Alexnet, DenseNet, EfficientNet, ResNet and different feature extraction methods including Mel Spectrogram, MFCC, LFCC on dog sound voice samples. For the comparative analysis dog-voice classes bark and howl are considered having total size of 103 samples, all samples having 16kHz sampling frequency. The DenseNet architecture recorded the best performance with an overall accuracy of 90% using Mel spectrograms feature. The ResNet family exhibited consistent performance across all feature extraction methods, achieving accuracy rates of approximately 86%.

Shah et al. [7] developed machine learning-based system that analyzes dog barks through spectrogram data and classifies them into emotional subtypes, including fear, aggression, and playfulness. A total of 320 dog bark samples were collected from multiple sources, including Kaggle and Pond5 open-source database, to ensure dataset diversity. Using Apple's Create ML, the model attained 93.2% accuracy, achieving 93%, 95%, and 97% for fear, aggression, and playfulness, respectively

Lekhak et al. [8] worked on DogSpeak a large-scale canine vocalization dataset containing 77,202 bark sequences collected from 156 dogs across five breeds. It is one of the most comprehensive datasets in animal vocalization research, offering detailed annotations such as breed, individual identity, and sex. The study introduced three benchmark tasks sex classification, breed classification, and individual dog recognition. Using HuBERT features with LR, RF, and XGB classifiers yielded the best performance for breed, sex, and individual recognition, respectively, achieving an overall macro F1-score and accuracy of approximately 52

Dang et al. [9] worked on dataset of 1,400 dog bark sequences with arousal and valence annotations—the largest available for Husky and Shiba Inu breeds—was introduced. For the Husky breed, the eGeMAPS feature set consistently outperformed other methods. In particular, eGeMAPS achieved the highest arousal classification accuracy of 53% using both LR and RF models, while valence classification reached 50% with RF and 51% with XGB.

Shivaprasad et al. [10] utilized hybrid filtering techniques Wavelet/LMS+IIR filter combined with MFCC-based feature extraction to enhance vocal signal quality. Emotion classes considered for the study included bark, growl and grunt with total recordings of 163 samples. For emotion classification, convolutional neural network architectures including AlexNet and LeNet-5 were evaluated. Experimental results demonstrated that AlexNet achieved a higher classification accuracy of approximately 87.5%, outperforming the LeNet-5 model, which attained around 85% accuracy.

Slobodian et al. [11] developed classifier with two convolutional layers with ReLU activation and a fully connected output layer with five neurons and softmax activation, with batch normalization applied throughout. Audio fragments of fixed length were processed using MFCC and Extractor Discovery System (EDS) features. The model was trained on 4,000 samples, validated on 800 records, and evaluated on a test set of 1,500 samples. Performance was assessed using precision, recall, F1-score, and accuracy, achieving an overall accuracy of approximately 72% when trained on data from a single dog.

Menon et al. [12] presented a notable advancement in Speech emotional Recognition modeling by explicitly incorporating multi-level attention within a CNN-BiLSTM hybrid architecture. It demonstrates the value of combining spectral feature learning with dynamic sequence modeling, guided by attention mechanisms that selectively emphasize emotionally salient information. Results demonstrated that the hybrid model consistently outperformed several state-of-the-art approaches on all three datasets. Reported average classification accuracies were 94.62% on Emo-DB, 67.85% on IEMOCAP and 95.80% on Amritaemo Arabic.

Tiwari et al. [13] presented a hybrid CNN-BiLSTM-Conv1D model that enhances SER by jointly modeling spectral and sequential speech features, achieving 94 % accuracy on RAVDESS and outperforming classical baselines. The approach demonstrates how integrating convolutional structures with bidirectional recurrent networks and 1D convolution for temporal refinement can strengthen the extraction of emotional cues in spoken audio, with practical deployment in an interactive web system.

Perez et al. [14] developed an automatic classifier that can identify individual dogs based on their barks. The author's created an dataset of 6000 barks by applying positive ans negative stimuli to dogs. Low level desciptors(LLD's) such as MFCC[0-14], Mel- Spectrogram and log power of mel frequency bands which showed high discriminative f1-measure were used for feature selection and SVM with polynomial kernel for classification. The resultant f1 measure(accuracy) obtained was 90.5%.

III. Proposed Method

This section deals with sequence of methods employed on to dataset in-order to improve the effectiveness of proposed model.

A. Dataset Preprocessing

The emotion dataset acquired using voice recorder app, underwent pre-processing methods comprising of resampling, slice or pad and normalization. The following schemes are incorporated in the system in-order to make data scale, dimension and feature invariant. Firstly, all the samples existing in the dataset are re-sampled to a sampling frequency of 16kHZ. The dimension(size) of the samples is let identical by the process of slicing and padding. Finally, the amplitudes of the voices are normalized using standard normalization technique.

1) Resampling: Voice samples acquired from various acquisition source are sampled at varying sampling rates, in order to maintain uniform sampling frequency between the voice samples frequency resampling is employed. Let x(n) represents discrete time signal with source sampling frequency f_s having N samples, and y(m) be the resampled sequence of length M with target sampling frequency f_t [15].

The value of M and y(m) is determined using the following equations (1) and (2)

$$M = \left\lfloor N \cdot \frac{f_t}{f_s} \right\rfloor \quad (1)$$

$$y\left[m\right] = \sum_{n=0}^{N-1} x\left[n\right]\ sinc\left(\frac{m}{f_t} - \frac{n}{f_s}\right) \quad (2)$$

2) Slice or Zero Padd: The objective of this method is to ensure that all voice samples have uniform length L. Let N represents length of discrete time sequence x(n). If $N > L$, the sequence x(n) is truncated to maintain length of L samples; otherwise, it is zero-padded to acheive the desired length of L samples.

The relationship between y(n) and x(n) as explained in the above section is given by equation (3)

$$y(n) = \begin{cases} x(n), & 0 \leq n < \min(N, L) \\ 0, & N \leq n < L \end{cases} \quad (3)$$

3) Peak Normalization: As, the voice samples are acquired from different acquisition system, the data-type of the voice samples are inconsistent. To maintain consistency in amplitude scale, the voice samples are normalized to the range [0,1].

The method used to normalize the data is given by the equation (4)

$$y[n] = \frac{x[n]}{\max\limits_{n} |x[n]|} \quad (4)$$

B. Addition of Real-time noise with varying noise levels (SNR) of 0dB, 5dB and 10dB

The pre-processed emotion vocalization is added with real time noise (Environment) like samples including Engine idling, Child playing, Street music, Jack Hammer. They are added to the voice samples in order to make the system robust for noise-like interference. These noise samples are selected from the UrbandSound8K which consists of 8000 audio samples belonging to 10 different voice classes. The superimposed signal with three different SNR levels (0,5 and 10 dB) produces the noisy emotion dataset.

Average Dog-vocalization and Environment noise power over 20 ms duration frame with N samples which is required to accurately control the noise injection process are mentioned in equations (5) and (6)

$$P_s = \frac{1}{N} \sum_{i=1}^{N} s^2(i) \quad (5)$$

$$P_n = \frac{1}{N} \sum_{i=1}^{N} n^2(i) \quad (6)$$

Noise scaling factor that adjusts the noise amplitude to achieve a predefined signal-to-noise ratio (SNR) is mentioned in equation (7)

$$\alpha = \sqrt{\frac{P_s}{P_n \cdot 10^{\mathrm{SNR}/10}}} \quad (7)$$

Using the computed scaling factor, the noisy signal is generated by superimposing scaled environmental noise onto the original dog vocalization as mentioned in equation (8)

$$x_{\text{noisy}}\left(i\right) = s\left(i\right) + \alpha \cdot n\left(i\right) \quad (8)$$

C. Denoising or Signal Enhancement

Wiener filter gain, which adaptively balances noise suppression and signal preservation in the frequency domain is addressed in equation (9)

$$G\left(k\right) = \frac{P_s\left(k\right)}{P_s\left(k\right) + P_n\left(k\right)} \quad (9)$$

The enhanced speech spectrum is obtained by applying the Wiener gain function[16] to the noisy signal spectrum as indicated in equation(10)

$$\hat{S}\left(k\right) = G\left(k\right) \cdot X\left(k\right) \quad (10)$$

D. Feature extraction

Features are extracted from the dog vocalized clean, noisy and enhanced emotion samples. The techniques adapted for extracting the features from the voice samples was achieved through Log Mel- Spectrogram [17].

The Mel spectrogram is obtained by projecting the power spectrum onto a bank of perceptually motivated Mel-scale filters, equation (11) addresses the same

$$M\left(m, t\right) = \sum_{k} \left|X\left(k, t\right)\right|^2 H_m\left(k\right) \quad (11)$$

Hm(k) represents the weight assigned to the k-th FFT frequency bin by the m-th Mel filter in the Mel filterbank and X(k,t) represents the complex-valued STFT coefficient at frequency bin kand time frame t.

To enhance feature robustness through the reduction of dynamic range variations, a logarithmic compression method is employed as described in equation (12).

$$\mathrm{LogMel}\left(m, t\right) = \log\left(M\left(m, t\right) + \varepsilon\right) \quad (12)$$

E. Classification

The three sets of features extracted emotion vocalizations act as input to the classifier. CNN+Bi-LSTM+Attention(Hybrid- DNN with Relationship between present and time-delayed samples) classifier is to perform classification. Performance metrics including the overall accuracy, individual class accuracy, Confusion matrix, Precision, Recall, F1-score and ROC graph are recorded for deciding the effectiveness of the proposed system against the state of art methods available. The model equations related to the Classifier is discussed detail in section (IV). The block diagram of the proposed system is shown below in the Fig.1.

The Pseudo-code of the proposed system is presented in Algorithm 1.

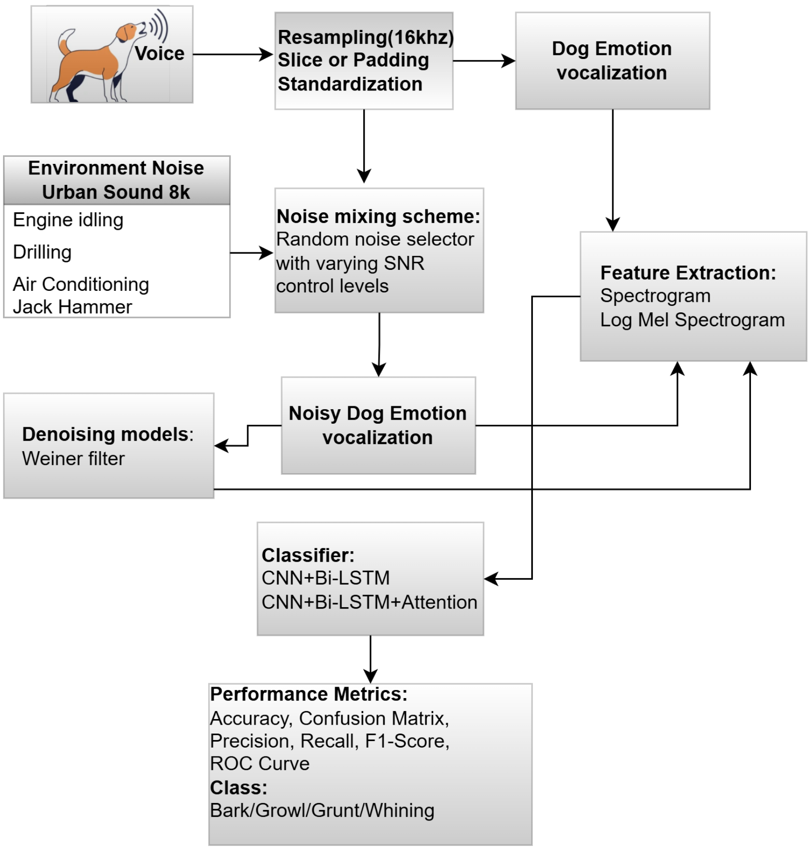

Fig. 1. Block diagram of the Robust Canine Denoising and Classification system

Algorithm 1 Pseudo Code of Proposed system

Require: Path to Dog-Emotion Dataset D
Require: Path to Environmental Noise Dataset N
Ensure: Classification accuracy and performance metrics
for each dog vocalization $d \in D$ **do**
  Preprocess d
    Resample to 16 kHz
    Maintain uniform signal length
    Normalize amplitude
  Extract acoustic features from d
  Store clean features with class labels
  Randomly select noise class $n \in N$
  Extract noise segment n_{seg} with same length as d
  for each SNR level $s \in \{0, 5, 10\}$ dB **do**
    Compute signal power P_d
    Compute noise power P_n
    Scale n_{seg} to satisfy SNR $= s$
    Generate noisy signal:
      $d_{noisy} \leftarrow d + \text{scaled}(n_{seg})$
    Extract features from d_{noisy}
    Store noisy features with class labels
    Apply denoising algorithm:
      $d_{denoised} \leftarrow \text{DENOISE}(d_{noisy})$
    Extract features from $d_{denoised}$
    Store denoised features with class labels
  end for
end for
Split dataset into training and testing sets
Train deep learning classifier (CNN-BiLSTM / CNN-BiLSTM-Attention)
Evaluate performance under:
  Clean condition
  Noisy condition (0, 5, 10 dB)
  Denoised condition (0, 5, 10 dB)
return Classification accuracy and performance metrics

IV. Hybrid Classifier

The block diagram of the hybrid Classifier, which includes mainly three modules CNN(Convolutional), BiLSTM(Bi-directional LSTM) and Self- Attention. Mel-Log Spectrogram, which is 2-Dimensional data serves as the input to the classification system is mentioned in Fig. 2 below. Convolution neural network are best suited to handle 2-D signals and are capable in obtaining the hidden latent variables from the 2-D data. The BiLSTM formulation captures long-term temporal dependencies by processing the feature sequence in both forward and backward directions. The concatenated hidden state encodes contextual information from past and future frames, which is crucial for emotion recognition in non-stationary dog vocalizations. The self-attention mechanism assigns adaptive importance weights to each temporal frame and aggregates them into a single context vector. This enables the model to focus on emotionally salient segments while suppressing less informative or noisy portions of the signal.

A. Convolution

The Log-Mel Spectrogram features in the form 2-D signal acts as input X, convolution operation that extracts local time–frequency patterns from the inputs are given in equation(13)

$$h_k^{(l)} = \sigma\left(W_k^{(l)} * X + b_k^{(l)}\right) \tag{13}$$

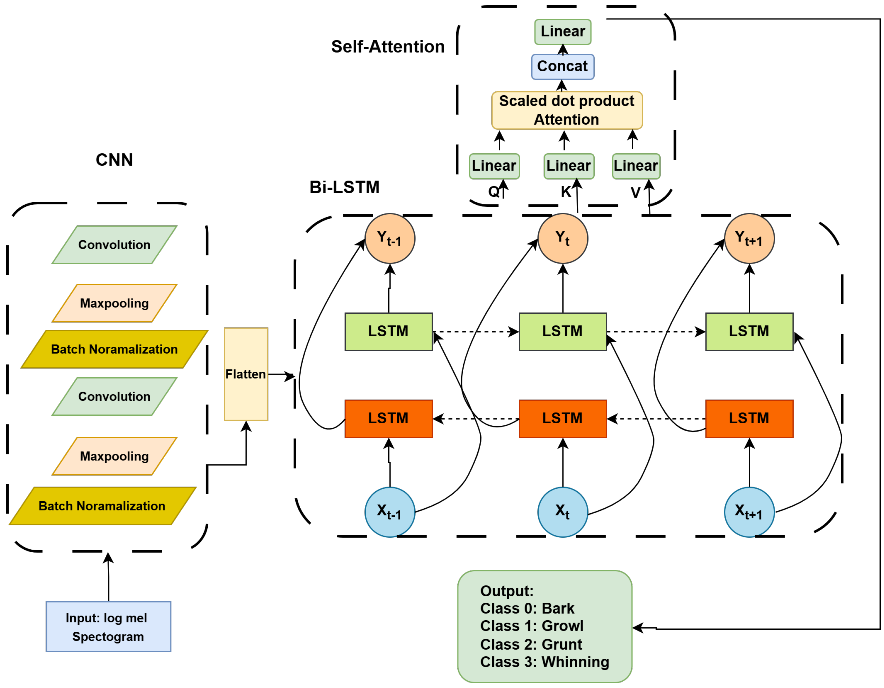

Fig. 2. Block diagram of the CNN-Bi LSTM-Attention Classifier

B. Max-pooling

Max-pooling (spatial down-sampling) is performed on to the convolutional feature maps to preserve dominant acoustic information

$$f_k^{(l)} = \max_{(i,j)\in\Omega} h_k^{(l)}(i,j) \tag{14}$$

C. Bi-LSTM layer

The down sampled CNN feature maps are transformed into temporal sequence suitable for recurrent processing using equation(15)

$$F = [f_1, f_2, \dots, f_T] \tag{15}$$

Temporal dependencies in dog vocalizations are detected by processing the feature sequence in the forward time direction is given by equation(16)

$$h_t^{(f)} = \mathrm{LSTM}\left(f_t, h_{t-1}^{(f)}\right) \tag{16}$$

Future contextual information is gained by processing the feature sequence in reverse order as shown in equation (17)

$$h_t^{(b)} = \mathrm{LSTM}\left(f_t, h_{t+1}^{(b)}\right) \tag{17}$$

The forward and backward hidden states of the feature vectors are aggregated to form a context-aware temporal representation as indicated in equation (18)

$$h_t = \left[h_t^{(f)}; h_t^{(b)}\right] \tag{18}$$

D. Sel-Attention

The scalar e_t represents the relevance score of the BiLSTM hidden state at time step t. It quantifies how informative a particular temporal frame is for emotion recognition by projecting the hidden representation into an attention space as observed in equation (19) [17]

$$e_t = w_a^\top \tanh(h_t) \tag{19}$$

The normalized weight α_t indicates the relative importance of the t-th frame within the entire sequence. By applying a softmax function, all attention weights sum to one, allowing the model to selectively emphasize emotionally salient frames is provided by equation (20)

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T}\exp(e_k)} \tag{20}$$

The context vector **c** is a weighted sum of the BiLSTM hidden states, where the weights are given by α_t. It provides a compact and discriminative representation of the temporal sequence by aggregating the most relevant emotional information across time. The equation describing the concept is highlighted below in equation (21)[18]

$$c = \sum_{t=1}^{T} \alpha_t h_t \tag{21}$$

V. Results and Discussion

A. System Requirements

All experiments were carried out on a system powered by a 12th-generation Intel® Core™ i5-1235U processor with a base clock speed of 1.30 GHz, supported by 16 GB of RAM. The machine operates on a 64-bit operating system and utilizes an x64-based processor architecture. The software environment was implemented using Python Interpreter version 3.11.9. Machine learning and deep learning experiments were executed using scikit-learn version 1.8.0 for classical machine learning utilities and TensorFlow version 2.18.0 for developing and training the proposed CNN–BiLSTM–Attention model. Fig.3 shows the CNN-BiLSTM-Attention model summary generated using Python-Tensorflow framework. Overall 12,48,836 total and trainable parameters were used while training.

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 256, 256, 1)	0	-
conv2d (Conv2D)	(None, 254, 254, 32)	320	input_layer[0][0]
max_pooling2d (MaxPooling2D)	(None, 127, 127, 32)	0	conv2d[0][0]
conv2d_1 (Conv2D)	(None, 125, 125, 64)	18,496	max_pooling2d[0][0]
max_pooling2d_1 (MaxPooling2D)	(None, 62, 62, 64)	0	conv2d_1[0][0]
conv2d_2 (Conv2D)	(None, 60, 60, 128)	73,856	max_pooling2d_1[0][0]
max_pooling2d_2 (MaxPooling2D)	(None, 30, 30, 128)	0	conv2d_2[0][0]
reshape (Reshape)	(None, 30, 3840)	0	max_pooling2d_2[0][0]
time_distributed (TimeDistributed)	(None, 30, 256)	983,296	reshape[0][0]
bidirectional (Bidirectional)	(None, 30, 128)	164,352	time_distributed[0][0]
attention (Attention)	(None, 30, 128)	0	bidirectional[0][0], bidirectional[0][0]
add (Add)	(None, 30, 128)	0	bidirectional[0][0], attention[0][0]
global_average_pooling1d (GlobalAveragePooling1D)	(None, 128)	0	add[0][0]
dense_1 (Dense)	(None, 64)	8,256	global_average_pooling1…
dropout (Dropout)	(None, 64)	0	dense_1[0][0]
dense_2 (Dense)	(None, 4)	260	dropout[0][0]

Fig. 3. Tensorflow based CNN-Bi LSTM-Attention model summary

B. Dog- Emotion Dataset-1

To recognize and classify the dog emotion voices effectively, four classes of the emotions are selected mainly: bark, growl, grunt and whining. Normal bark voices are produced when the dog encounters are stranger. Growling voices are produced when the dog is very angry or involved with an fight with another dog. Grunting voices are produced when the dog feels discomfort. Whining voices are produced either when the dog wants to be out or hungry. When the dog wants to go out, it produces low whining tone and while hungry it produces high whining tones. Table 1 provides summary about the various types of dog vocalization and corresponding contexts with detailed descriptions.

The size of the dataset considered for the research work is 239 samples. Individual bark, growl, grunt and whining class voice samples size were 54, 60, 80 and 45 respectively. Majority of the dog-voices were acquired from the voice recorder app, sampled to 44.1 kHz and the rest were sampled at 16 kHz [20]. Fig. 4-7, illustrates the Mel-spectrogram representation of various dog emotion vocalizations[25]. Bark and Growl voices exhibit dominant low-frequency energy with sustained temporal patterns, while grunt sounds appear as short-duration broadband events. In contrast, whining vocalizations demonstrate higher-frequency tonal components with smoother temporal evolution. These observations confirm that Mel-spectrograms effectively capture discriminative time–frequency characteristics of different dog vocalization types.

TABLE 1
TYPES OF DOG VOCALIZATIONS AND ASSOCIATED CONTEXT

Type of Voice	Context	Description
Normal Bark	Stranger	Produced when the dog encounters unfamiliar people and reacts to their presence
Growling	Angry / Fighting	Occurs when the dog is aggressive or involved in a fight with another dog
Grunting	Discomfort	Indicates uneasiness, pain, or physical discomfort in the dog
Whining (Low Tone)	Wants to go out	Low-pitched whining sound when the dog wants to go outside
Whining (High Tone)	Hungry	High-pitched whining sound when the dog is hungry

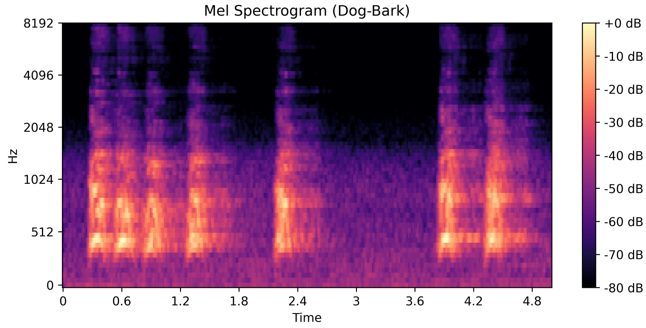

Fig. 4. Mel Spectrogram of dog-bark voice

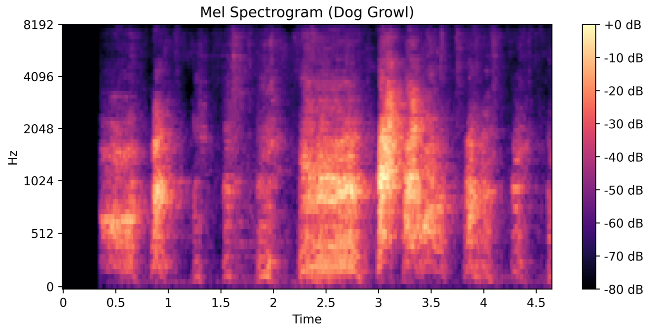

Fig. 5. Mel Spectrogram of dog-growl voice

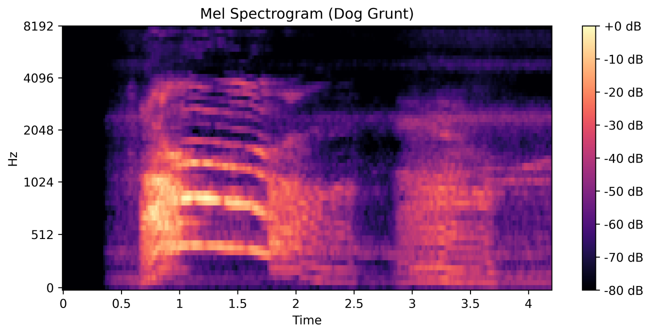

Fig. 6. Mel Spectrogram of dog-grunt voice

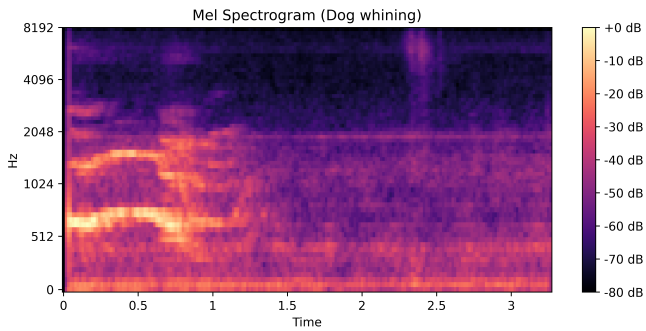

Fig. 7. Mel Spectrogram of dog-whining voice

1) Accuracy and Loss functions using various Deep Learning Models on Clean Dog-emotion dataset-1: Table 2 presents the comparative results obtained on Dog Emotion Dataset-1 using various state-of-the-art deep learning classifiers. The models evaluated include CNN, LeNet-5, AlexNet, CNN-LSTM, VGGNet, CNN–Transformer Encoder, CNN–BiLSTM, and the proposed CNN–BiLSTM with Attention. The system uses clean dog vocalizations as input, comprising 180 samples across four emotion classes. Feature extraction is performed using spectrograms and log-Mel spectrogram representations. Under controlled conditions, CNN–BiLSTM–Attention and CNN–LSTM (log-Mel features) attained 100% training accuracy, with validation accuracies of 91.67% and 89.58%, respectively.

2) Accuracy and Loss functions using various Deep Learning Models on Noisy Dog-emotion dataset-1: Data Augmentation: To measure the robustness of dataset on various classifier, the dataset underwent data-augmentation method. In the data-augmentation approach, each data was added with real time noise of varying noise levels 0dB, 5dB and 10dB. Total augmented data created is three times original dataset size, approximately 717 voice samples.

The dataset was split into training (80% of 717 samples) and validation (20% of 717 samples) sets. Table 3 presents the training accuracy, training loss, validation accuracy, and validation loss obtained using various state-of-the-art classifiers, along with the proposed model, based on spectrogram and log-Mel spectrogram features. The proposed CNN–BiLSTM–Attention model and the AlexNet-based CNN architecture achieved superior performance when using log-Mel spectrogram features. A training accuracy of 100% and a validation accuracy of 99.31% were recorded.

3) Accuracy and Loss functions using various Deep Learning Models on Denoised Dog-emotion dataset-1: Weiner Denoising: To filter the noisy dog vocalization, frequency domain scheme-based wiener filtering is adapted. The denoised dog-vocalization serves as the input to the emotion recognition system and produces spectrogram or log-mel spectrogram features as discussed in the previous section. These features serves as the input to the state-of-art classifiers. The proposed classifier (CNN-BiLSTM-Attention), CNN-LSTM and ALexnet CNN produced great results on log-mel spectrogram features. The training and validation accuracy recorded using the proposed classifier was 100% and 97.2% as indicated in Table 4.

4) Confusion matrix and Individual Class Accuracy on Denoised dog emotion dataset-1: Fig. 8 shows the confusion matrix generated using CNN-BiLSTM-Attention classifier on the denoised dog-vocalization dataset. Total number of supports used for validation is 143 voice samples. Dog-growl, Dog-grunt and Dog-whining class distributions are 33, 36, 48 and 26 samples respectively. The result of correct classification for bark, growl, grunt and whinning classes were 33, 36, 45 and 25. Moreover, the class accuracy for each of the classes are determined and displayed in the form of bar-graph as shown in Fig. 9.

5) Classification report and ROC curve on Denoised dog emotion dataset-1: The details about the precision, recall, f1-score, macro average are derived from the confusion matrix chart discussed in the Fig. 10. The precision score for the dog-grunt class is evaluated by dividing the number of correct dog-grunt predictions (45) by all dog-growl predictions (46), which is 98%. Similarly, recall score for the dog-grunt class is given by dividing total number of correct dog-grunt predictions (45) by all dog-grunt instances (48), which is 96%. In a similar way, the dog grunt and dog whining emotion precision and recall values are calculated. Using the precision and recall value for each of the class, the f1-

TABLE 2
TRAINING ACCURACY, VALIDATION ACCURACY, TRAINING LOSS, AND VALIDATION LOSS FOR DOG EMOTION DATASET-1 WITH CLEAN DOG VOCALIZATIONS

	Spectrogram (Epoch = 30)				Log Mel-Spectrogram (Epoch = 30)			
Classifier	Train Acc (%)	Val Acc (%)	Train Loss	Val Loss	Train Acc (%)	Val Acc (%)	Train Loss	Val Loss
CNN	99.47	85.42	0.0164	1.8680	99.47	87.50	0.0097	1.1052
LeNet-5	100	87.50	1.44×10^{-4}	1.8486	100	89.58	1.17×10^{-4}	1.1363
AlexNet	87.89	77.08	0.2954	0.8969	97.37	83.33	0.0792	1.0490
CNN–LSTM	83.68	81.25	0.3860	0.6109	100	89.58	0.0021	0.6095
VGGNet	80.00	85.42	0.5198	0.4155	76.32	75.00	0.5627	0.6367
CNN–Transformer Encoder	80.00	72.92	0.4317	0.7196	84.21	79.17	0.3644	0.5881
CNN–BiLSTM	87.89	75.00	0.2707	0.8134	93.68	81.25	0.1540	0.6468
CNN–BiLSTM–Attention (Proposed)	**90.53**	**81.25**	**0.2489**	**0.6613**	**100**	**91.67**	**0.0261**	**0.6103**

TABLE 3
TRAINING ACCURACY, VALIDATION ACCURACY, TRAINING LOSS, AND VALIDATION LOSS FOR DOG EMOTION DATASET-2 WITH NOISY DOG VOCALIZATIONS

	Spectrogram (Epoch = 30)				Log Mel-Spectrogram (Epoch = 30)			
Classifier	Train Acc (%)	Val Acc (%)	Train Loss	Val Loss	Train Acc (%)	Val Acc (%)	Train Loss	Val Loss
CNN	86.51	91.61	0.3877	0.3378	96.85	97.20	0.0913	0.0756
LeNet-5	100	97.90	6.29×10^{-5}	0.2181	100	93.71	2.89×10^{-5}	0.1993
AlexNet	76.18	72.73	0.5419	0.5575	100	99.31	5.81×10^{-4}	0.1174
CNN–LSTM	76.36	71.33	0.5966	0.6939	96.85	94.41	0.0789	0.1199
VGGNet	73.65	70.63	0.6328	0.6353	90.72	88.81	0.2365	0.3428
CNN–Transformer Encoder	56.39	57.34	0.9675	0.9441	88.44	88.11	0.3149	0.2986
CNN–BiLSTM	99.65	90.21	0.0190	0.2463	99.82	95.10	0.0085	0.1517
CNN–BiLSTM–Attention (Proposed)	**92.12**	**86.01**	**0.5407**	**0.6140**	**100**	**99.30**	**0.0396**	**0.0723**

TABLE 4
TRAINING ACCURACY, VALIDATION ACCURACY, TRAINING LOSS, AND VALIDATION LOSS FOR DENOISED (WIENER FILTERING) DOG EMOTION VOCALIZATIONS

	Spectrogram (Epoch = 30)				Log Mel-Spectrogram (Epoch = 30)			
Classifier	Train Acc (%)	Val Acc (%)	Train Loss	Val Loss	Train Acc (%)	Val Acc (%)	Train Loss	Val Loss
CNN	87.74	88.81	0.5205	1.6767	96.50	97.90	0.0904	0.0935
LeNet-5	100	91.61	2.17×10^{-6}	0.4592	100	96.50	2.1×10^{-5}	0.1759
AlexNet	81.96	75.52	0.4756	0.6175	100	97.90	4.8×10^{-5}	0.3278
CNN LSTM	76.19	70.27	0.5929	0.7183	99.82	97.90	0.0169	0.0492
VGGNet	70.93	73.43	0.6645	0.7375	88.09	85.31	0.2440	0.3792
CNN–Transformer Encoder	68.13	73.43	0.7701	0.7047	90.89	93.01	0.2651	0.2155
CNN–BiLSTM	99.30	91.61	0.0432	0.3268	100	96.50	9.3×10^{-4}	0.1637
CNN–BiLSTM–Attention (Proposed)	**93.35**	**81.82**	**0.5043**	**0.6515**	**100**	**97.20**	**0.0400**	**0.1019**

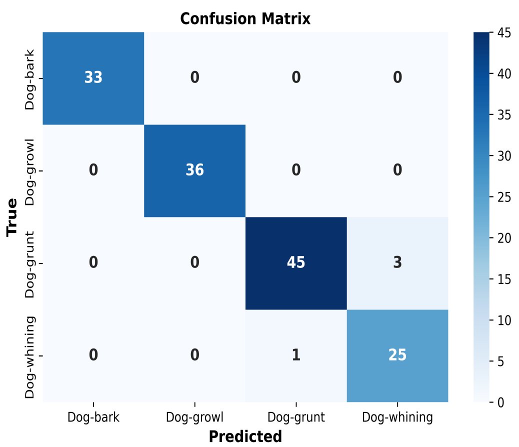

Fig. 8. Confusion matrix using CNN-BilSTM-Attention Classifier

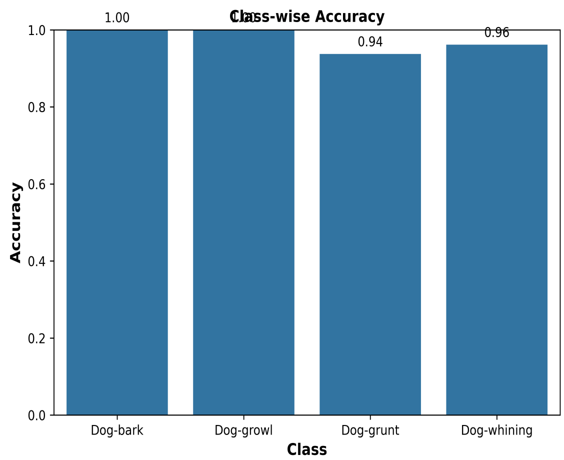

Fig. 9. Class-wise Accuracy using CNN-BilSTM-Attention Classifier

score can also be determined as they are connected with one another.

Fig. 11 Illustrates the multi-class ROC curves obtained using the one-vs-rest strategy[21]. The proposed

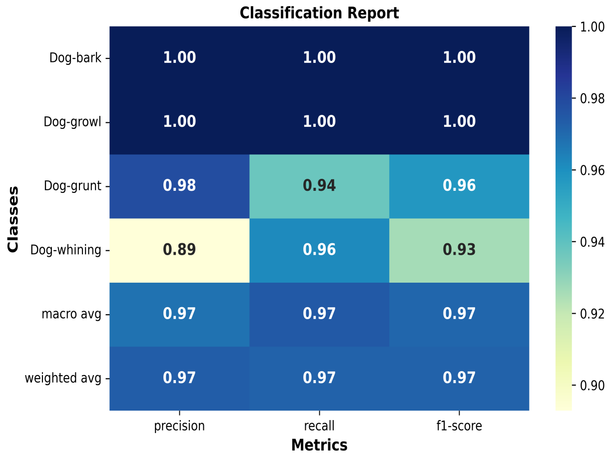

Fig. 10. Classification report using CNN-BiLSTM-Attention Classifier

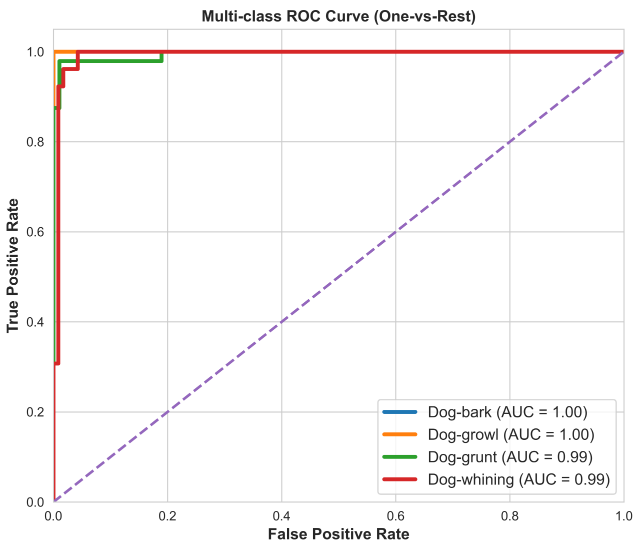

Fig. 11. Multi Class ROC Curve using CNN-BilSTM-Attention Classifier

CNN–BiLSTM–Attention model demonstrates strong discriminative capability, achieving AUC values of 1.00,1.00, 0.99, and 0.99 for the four dog-vocalization classes, respectively. The perfect AUC observed for dog-bark and dog-growl voice suggests that its acoustic patterns are highly distinctive, while the slightly lower AUC values for the remaining dog-grunt and dog-whining classes indicate minor overlap in the learned feature space. Overall, the results confirm the robustness and effectiveness of the proposed model in distinguishing between different classes.

C. Dog-Emotion dataset-2

Publicly available dataset to discriminate dog bark and dog howl voice is considered in this section. The dataset includes two classes, dog-bark and dog-howl voice samples, each sampled at sampling frequency of 16kHz. The size of the dataset was 103 samples. Dog bark (Class-1) samples were 46 in numbers and dog-howl(Class-2) voice samples were 57 in numbers [6], [22]. To determine the classifiers accuracy in discriminating the various classes, five best Deep Learning models namely CNN, Lenet-5, Alexnet, CNN-LSTM, CNN-BiLSTM and CNN-BiLSTM-Attention are selected based on performance while being applied on dog-emotion dataset-1. Table 4, Table 5 and Table 6 presents training accuracy, validation accuracy recorded for the clean, noisy and denoised dog-emotion dataset-2. The proposed classifier with log mel Spectrogram showed remarkable results of 100% Accuracy on the clean bark vs howl dataset. Also, the result recorded on the noisy and denoised dataset were higher than 98%. Table 6, Table 7 and Table 8 presents training accuracy, validation accuracy recorded for the clean, noisy and denoised dog-emotion dataset-2. The proposed classifier with log mel Spectrogram showed remarkable results of 100% Accuracy on the clean bark vs howl dataset. Also, the result recorded on the noisy and denoised dataset were higher than 98%.

TABLE 5
TRAINING AND VALIDATION ACCURACY FOR CLEAN DOG EMOTION VOCALIZATIONS

	Spectrogram		Log Mel-Spectrogram	
Classifier	Train Acc (%)	Val Acc (%)	Train Acc (%)	Val Acc (%)
CNN	97.56	90.48	100	95.24
LeNet-5	100	95.24	100	95.24
AlexNet	97.56	100	100	100
CNN–LSTM	100	95.24	100	100
CNN–BiLSTM	100	95.16	100	98.39
CNN–BiLSTM–Attn	**98.79**	**93.55**	**100**	**98.39**

TABLE 6
TRAINING AND VALIDATION ACCURACY FOR NOISY DOG EMOTION VOCALIZATIONS

	Spectrogram		Log Mel-Spectrogram	
Classifier	Train Acc (%)	Val Acc (%)	Train Acc (%)	Val Acc (%)
CNN	99.19	91.94	99.64	100
LeNet-5	100	93.55	100	96.77
AlexNet	85.83	83.87	100	100
CNN–LSTM	100	96.77	100	95.16
CNN–BiLSTM	99.6	91.94	100	96.77
CNN–BiLSTM–Attn	**98.79**	**91.94**	**99.19**	**100**

TABLE 7
TRAINING AND VALIDATION ACCURACY FOR DENOISED DOG EMOTION VOCALIZATIONS

	Spectrogram		Log Mel-Spectrogram	
Classifier	Train Acc (%)	Val Acc (%)	Train Acc (%)	Val Acc (%)
CNN	99.19	96.77	99.6	98.39
LeNet-5	100	95.16	100	96.77
AlexNet	89.07	87.1	100	100
CNN–LSTM	100	90.32	100	95.16
CNN–BiLSTM	100	95.16	100	98.39
CNN–BiLSTM-Attn	**98.79**	**93.55**	**100**	**98.39**

D. Comparison of Proposed method with state of art Deep Learning Classifiers

The sections discusses the comparison of accuracy score obtained by the proposed system with the state of art techniques employed for dog-emotion classification. Accuracy achieved in [5] on three different sets of databases using Synchronsqueezing STFT and Mamba model were 91.97%, 95.36% and 67.25%. Work in [6], used Mel Spectrogram for feature extraction and DenseNet model for classification thereby achieving an accuracy of 90% on publicly available

TABLE 8
COMPARISON OF EXISTING METHODS FOR DOG EMOTION AND VOCALIZATION ANALYSIS

Paper Title	Dataset / Classes	Feature Extraction / Noise Removal	Classifier	Accuracy
A Barking Emotion Recognition Method Based on Mamba and Synchrosqueezing STFT [5]	Self-constructed dataset, UrbanSound8K, IEMOCAP	Synchrosqueezing STFT	Mamba model	91.97%, 95.36%, 67.25%
Voice Analysis in Dogs with Deep Learning [6]	103 samples; 2 classes (bark, howl)	Mel spectrogram	DenseNet	90%
Acoustic Classification of Canine Vocalizations Using Spectrogram-Based ML [7]	320 samples; 6 emotional states	Spectrogram	Custom ML model	93.2%
DogSpeak: A Canine Vocalization Classification Dataset [8]	77,202 samples; multi-task (sex, breed, individual)	HuBERT features	LR, RF, XGBoost	52%
Dog Emotion Identification Using Hybrid Filter Models [10]	163 samples; 4 classes	MFCC, IIR, Wavelet, LMS	AlexNet, LeNet-5	87.5%
ML-Based Animal Emotion Classification Using Audio Signals [11]	6300 samples; 6 classes	MFCC, EDS	2-layer CNN	72%
Automatic Individual Dog Recognition Based on Bark Acoustics [14]	6000 samples	MFCC, Mel spectrogram	SVM (polynomial kernel)	90.5%
Polyphony of Domestic Dog Whines and Vocal Cues to Body Size [23]	824 samples; 20 dogs	Spectrogram analysis, correlation measures	ANOVA	Negative correlation observed
Classification of Dog Barks: A Machine Learning Approach [24]	6646 samples; 6 contexts	Spectral features	ANOVA, Bayes	43% (noisy), 52% (controlled)
Proposed Hybrid Classifier	**239 samples; 4 classes (clean, noisy, denoised)**	**Log-mel spectrogram**	**CNN–BiLSTM–Attention**	**92%(Clean), 99%(Noisy), 97%(Denoised)**

bark vs howl dataset. Authors in [10], used MFCC feature extraction method and Alexnet Classication model on the self-created dataset thereby achieving an accuracy of 87%. The work in [11] with MFCC and EDS features and CNN classifier produced accuracy score of 72%. The work in [24] used Formant, STD of harmonics, RMS of spectral skewness features with Anova and Bayes classifier yielding an accuracy score of 52%. The proposed system used two datasets, self created dataset and publicly available dataset. The Classification results on both the datasets were above 90%. The complete summary report between various dog-emotion classification and proposed model is shown in Table 8.

VI. CONCLUSION

The paper proposed Hybrid Deep Learning classifier for effective classification of dog voices. To preserve the robustness of the classification report, data augmentation scheme is employed with the help of real time noises which are added to the voice samples randomly. The classifier proved to be efficient in handling and classifying the dog voice samples without noise, with noise and denoised conditions. The validation accuracy recorded on these conditions were observed to be 92%, 99% and 97% respectively. The objectives outlined in the introduction, which include developing an efficient voice recognition and classification model, conducting a comparative analysis of feature extraction methods, comparing the proposed classifier against state-of-the-art deep learning classifiers, and assessing the classifier's robustness by incorporating real-time noise, have all been thoroughly addressed.

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